

Spatiotemporal Aliasing in Video Action Recognition: A Cross-Architecture Analysis at Scale

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Paper ID #*****

Abstract. Temporal resolution in video action recognition is usually treated as a fixed hyperparameter rather than a sampling constraint. We present a large-scale study of spatiotemporal aliasing across 8 architectures and 8 datasets (8,000+ evaluation configurations). Aliasing is strongly associated with *attention mechanism type*: VideoMamba and TimeSformer alias $3\text{--}5\times$ less than CNNs and ViViT, despite ViViT being a Transformer. We introduce the **Temporal Demand Score (TDS)**, an architecture-averaged accuracy drop under temporal subsampling. FineGym (55.9pp) rivals sign language (AUTSL, 53.0pp), establishing fine-grained sport as a high-aliasing regime. Cross-resolution evaluation shows CNNs degrade up to 73pp off-native, while Transformers stay within $\pm 8\text{pp}$ on 7 of 8 datasets; target-resolution fine-tuning recovers up to 39pp for CNNs. Measured latency identifies VideoMAE as the most efficient Transformer (1.6–2.8ms) and VideoMamba as $4\text{--}8\times$ slower (8–15ms). Finally, entropy routing reduces average inference cost by 52% with no accuracy penalty, outperforming FrameExit [9] by +4.2pp.

Keywords: Temporal aliasing · Video recognition · Routing

1 Introduction

Applications of activity recognition include driving, healthcare, and surveillance [2, 13]. The field has evolved from 3D ConvNets and dual-pathway networks [4, 8, 29, 31, 35] to video Transformers and state-space models [1, 3, 7, 15, 18, 27, 33].

Despite this diversity, all models are evaluated using standardized sampling recipes (8–32 uniformly-sampled frames at fixed stride), treating temporal resolution as a static implementation detail. This assumption fails in resource-constrained deployments where dense sampling is infeasible [17], and leaves two fundamental questions unanswered: *when* does temporal subsampling become destructive, and *why* do different architectures respond differently?

Sampling theory perspective. The Nyquist–Shannon theorem [23] states that a signal must be sampled above twice its highest-frequency component to avoid aliasing, a principle also relevant to perception [32] and deep learning [38]. Human actions have dataset- and class-dependent temporal frequencies; sparse

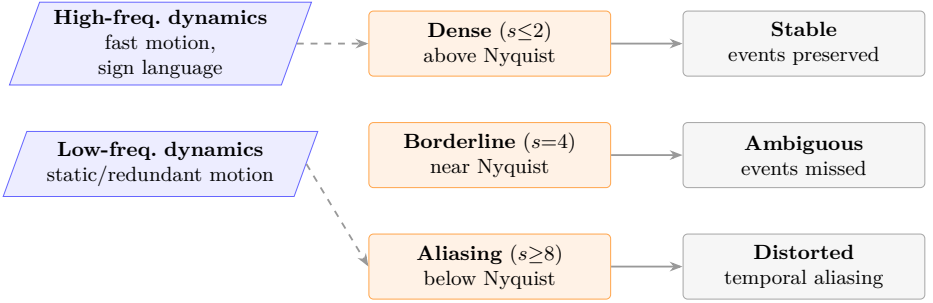


Fig. 1: Temporal aliasing: high-frequency actions enter the distorted-evidence regime when sampling stride exceeds the Nyquist limit for that action class.

video sampling can therefore distort rather than merely compress the evidence available to a recognizer.

Contributions. We conduct a systematic cross-architecture study of spatiotemporal aliasing:

1. **Temporal Demand Score (TDS):** an architecture-averaged dataset metric that consistently ranks 8 datasets by aliasing sensitivity, validated by leave-one-architecture-out analysis, architecture-family ablation (CNN-only, Transformer-only), and bootstrap confidence intervals (Sec. 4.1; supplementary material). FineGym (55.9pp) rivals sign language (AUTSL, 53.0pp) as the highest-demand domain.
2. **Cross-architecture aliasing characterization:** 8,000+ evaluation configurations reveal that *attention mechanism type*—not architecture family—is the strongest correlate of aliasing behavior, with VideoMamba and TimeSformer aliasing $3\text{--}5\times$ less than CNNs and ViViT (Sec. 4.2).
3. **Target-resolution fine-tuning:** Fine-tuning each model at the deployment resolution closes the cross-resolution gap by up to 39pp for CNNs while Transformers—already resolution-robust—require no retraining (Sec. 4.5).
4. **Measured inference latency:** GPU latency characterization across all 8 models $\times$ 5 resolutions (bfloat16), enabling Nyquist-aware deployment budgeting per device class (Sec. 4.6).
5. **Practical illustration and reproducibility package:** as a low-cost downstream use of the temporal-sweep results above—rather than a new routing algorithm in itself—a simple entropy-threshold router reduces average inference frames by 52% while matching dense-sampling accuracy, outperforming FrameExit by +4.2pp under a matched-backbone comparison. We also provide an anonymized supplementary package containing the interactive dashboard, evaluation scripts, and result tables; the public dashboard and repository will be released after review (Sec. 4.7).

2 Related Work

2.1 Video Recognition Architectures

CNN-based architectures. 3D ConvNets learn spatiotemporal features end-to-end [4, 29], while Non-local Networks introduce global self-attention [35]. Factorized variants such as R(2+1)D and MC3 improve optimization and efficiency [30, 31]; SlowFast processes two frame-rate streams simultaneously [8].

Transformer and SSM architectures. Video Transformers extend patch-based recognition to time: TimeSformer uses divided space-time attention [3], ViViT uses factorized attention [1], Video Swin uses local windows [18], MViT uses multi-scale features [7], and VideoMAE learns via masked patch reconstruction [27, 33]. VideoMamba [15] brings selective state-space scanning [6, 11] to video. Prior robustness work benchmarks spatial and photometric corruptions [21]; we instead isolate temporal and spatial sampling effects across this architectural spectrum.

2.2 Temporal Sampling and Adaptive Inference

Efficient inference methods sample or process fewer frames. TSN and TSM define sparse-sampling recipes [17, 34]; AdaFrame selects frames [37]; AdaFocus crops regions [36]; AR-Net adapts resolution [20]; and FrameExit uses confidence cascades [9]. Dataset-level temporal reasoning has been studied with shuffled frames [12] and motion-only cues [22]. These works do not measure the accuracy cliff induced by controlled stride and coverage sweeps; our TDS metric provides an architecture-averaged temporal-demand ranking.

2.3 Sampling Theory in Deep Learning

Sampling theory predicts that undersampling a signal below its highest temporal frequency creates aliasing [23]. Temporal aliasing is well documented in perception [32]; in deep learning, strided operators can violate shift-invariance unless anti-aliased [38]. Frequency-domain analysis has been explored for action recognition [28], but not as a large-scale cross-architecture stress test of modern video models.

3 Methodology

3.1 Datasets and Architectures

Table 1 summarizes the 8 datasets selected to span semantic domains, motion frequencies, and recording conditions. We evaluate 8 architectures from four families: (i) **3D CNNs**: R3D-18, MC3-18 [30], R(2+1)D-18 [31]; (ii) **dual-pathway**: SlowFast-R50 [8]; (iii) **Transformers**: TimeSformer [3], ViViT [1], VideoMAE [27]; (iv) **SSM**: VideoMamba [15]. All models are fine-tuned from Kinetics-400 checkpoints for 10 epochs using AdamW with cosine decay; learning rate 2×10^{-5} (Transformers/SSMs) and 10^{-4} (CNNs).

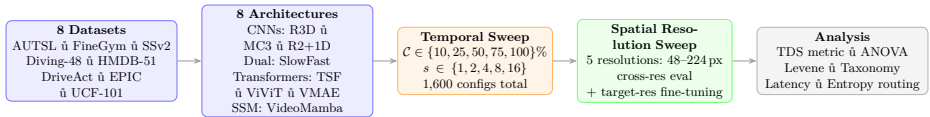


Fig. 2: Pipeline: 8 architectures evaluated across 8 datasets $\times$ 5 resolutions $\times$ 5 coverages $\times$ 5 strides = **8,000+ configurations**. Temporal sweeps feed TDS, ANOVA, Levene, and entropy routing; spatial sweeps characterize cross-resolution robustness and target-resolution fine-tuning.

Table 1: TDS = mean accuracy drop (stride=1 $\rightarrow$ 16, cov=100%) per Eq. 1, averaged over all 8 architectures ($n=8$). Sorted by TDS. **Bold** = the two co-equal highest-demand domains (their relative order is leave-one-out sensitive; ranks 3-8 are stable under the supplementary robustness analysis).

Dataset	Classes	Videos	Domain	TDS	n
FineGym [24]	97	1,960	Fine-grained sport	55.9pp	8
AUTSL [25]	226	4,418	Sign language	53.0pp	8
SSv2 [10]	174	3,464	Causal/temporal	27.6pp	8
DriveAct [19]	33	448	In-vehicle	21.9pp	8
Diving-48 [16]	48	743	Fine-grained	19.2pp	8
HMDB-51 [14]	51	885	Sports	16.6pp	8
EPIC-Kitchens [5]	89	1,020	Egocentric	9.7pp	8
UCF-101 [26]	101	2,020	Appearance	4.9pp	8

3.2 Temporal Sampling Protocol

We decouple *temporal coverage* $\mathcal{C}$ (fraction of clip observed) from *temporal stride* s (sampling density), forming a 5×5 grid: $\mathcal{C} \in \{10, 25, 50, 75, 100\}\%$ and $s \in \{1, 2, 4, 8, 16\}$. The temporal sweep at native resolution yields $8 \times 8 \times 25 = \mathbf{1,600}$ model-dataset-configuration triples (each evaluated over ≈ 20 clips per class). Combined with the spatial resolution sweep (5 resolutions), the total reaches **8,000+ evaluation configurations**.

Coverage implementation. Coverage is applied from the *beginning* of each clip: the first $\lfloor T \cdot \mathcal{C}/100 \rfloor$ frames form the candidate pool, and s controls sampling density within that window. This simulates a sensor that observes only a leading fraction of the action and gives the sharpest measurement of how early the model has enough temporal context. Random window placement would confound temporal structure with missing-data effects; a 10% random-window check on SSv2 gives qualitatively similar coverage effects (supplementary material).

3.3 Spatial Resolution Protocol

Cross-resolution evaluation. We evaluate all 8 native-resolution checkpoints on all 8 datasets at 48, 96, 112, 160, 224, and 336 px without retraining, measuring accuracy retention when the input resolution differs from the native training resolution (112px for CNNs, 224px for Transformers and VideoMamba).

Target-resolution fine-tuning. For each model-dataset-resolution triple at 48–224px, we fine-tune from the native-resolution checkpoint for 10 additional epochs. For Transformers and VideoMamba, positional embeddings are bicubically interpolated from the native grid to the target grid before fine-tuning; discarding mismatched positional embeddings causes a large accuracy cliff. CNNs transfer directly, and SlowFast uses adaptive average pooling to support non-native resolutions.

3.4 Temporal Demand Score (TDS)

We define TDS as the architecture-averaged accuracy drop from stride=1 to stride=16:

$$\text{TDS}(\mathcal{D}) = \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} [\text{Acc}(m, \mathcal{D}, s=1) - \text{Acc}(m, \mathcal{D}, s=16)]^+ \quad (1)$$

where $[\cdot]^+$ excludes feature-collapsed models. TDS is architecture-pool averaged, but its dataset ranking is stable under leave-one-architecture-out, family ablations, and bootstrap resampling over our 8-model pool (Sec. 4.1; supplementary material), suggesting it reflects dataset temporal structure rather than any single model.

3.5 Statistical Analysis

For each model-dataset pair, two-way ANOVA decomposes accuracy variance into coverage and stride components (effect size η^2). Levene’s test detects per-class variance inflation under sparse sampling. Post-hoc Welch tests with Bonferroni correction identify aliasing cliffs.

3.6 Entropy-Based Routing

Given temporal-sweep results at two operating points—cheap: cov=100%/s=4 (4 effective frames); dense: cov=100%/s=1 (16 frames)—we route video v by threshold:

$$\hat{y}(v) = \begin{cases} \hat{y}_{\text{cheap}} & \text{if } \max_c p(c \mid v, s=4) > \tau \\ \hat{y}_{\text{dense}} & \text{otherwise} \end{cases}$$

We sweep $\tau \in [0.10, 0.95]$ and report the Pareto-optimal point at avg frames ≤ 8 . No retraining is required; only the existing temporal-sweep inference is reused.

4 Results

4.1 Temporal Demand Score Validates Architecture-Independent Ranking

Figure 3 (left) ranks datasets by TDS. **FineGym (55.9pp)** is the most important new finding in this dimension: fine-grained gymnastics phase transitions

Table 2: Spectral validation: Pearson r between optical-flow magnitude and per-class aliasing loss. Flow alone partially explains temporal demand; camera-motion-heavy datasets (EPIC, UCF) show weak correlation.

Dataset	Classes	Pearson r	Significant	Interpretation
FineGym	97	-0.475	✓	Discrete-event classes alias more
DriveAct	33	0.327	marginal	Flow $\sim$ arm motion
AUTSL	226	0.285	✓	Flow $\sim$ hand speed
SSv2	174	0.184	✓	Causal/object motion
HMDB-51	51	0.180	×	Mixed motion types
Diving-48	47	0.140	×	Body rotation dominated
UCF-101	101	0.031	×	Appearance dominates
EPIC-Kitchens	89	-0.002	×	Camera motion confounds

rival AUTSL sign language (53.0pp), showing that sport is not inherently low-TDS when the label space requires sub-second phase discrimination. AUTSL requires dense sampling to discriminate fine handshape transitions, but FineGym’s phase boundaries are comparably demanding. UCF-101 (4.9pp) is nearly appearance-driven: object identity and static scene cues often suffice.

Crucially, ranks 3–8 are stable under leave-one-architecture-out: removing any single architecture leaves SSv2, DriveAct, Diving-48, HMDB-51, EPIC-Kitchens, and UCF-101 in the same order. Only the #1/#2 position swaps when VideoMamba is removed, so we report FineGym and AUTSL as co-equal high-TDS domains. This stability also holds for CNN-only, Transformer-only, and Transformer+SSM pools (Spearman $\rho \geq 0.976$ against the full pool), while bootstrap resampling gives a FineGym–AUTSL 95% CI of $[-4.6, +11.7]$ pp, consistent with zero (supplementary material).

Spectral validation. Figure 3 (right) correlates optical-flow magnitude with per-class aliasing loss. AUTSL and DriveAct show the expected positive within-dataset trend: classes with more motion alias more. EPIC-Kitchens is near zero, suggesting that egocentric camera motion decouples raw flow from action-relevant demand. FineGym is inverted ($r = -0.475$, $p < 0.0001$): high-flow gymnastics classes often remain visually distinctive from almost any frame, while low-flow held positions or balance corrections depend on brief key moments. Thus raw flow magnitude is only a partial proxy for temporal demand, especially when class identity hinges on a discrete event rather than sustained motion (Table 2).

The inversion generalizes beyond FineGym. A direct spectral test on all 8 datasets across 5 resolutions pairs whole-frame optical-flow cutoff frequency with empirical stride-sensitivity at the same resolution (40 points; supplementary material). The relationship is significantly *negative* ($\rho = -0.63$, $p = 1.1 \times 10^{-5}$): UCF-101 and HMDB-51 have high flow-derived cutoffs but low stride-sensitivity, while FineGym and AUTSL have the highest stride-sensitivity at lower cutoff frequencies. Thus whole-frame flow does not literally validate the Nyquist analogy; it mainly shows that classifier-relevant temporal evidence is often localized and diluted by bulk motion.

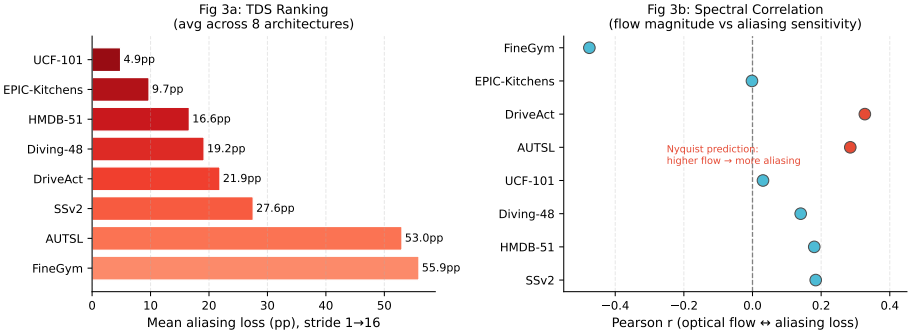


Fig. 3: Left: TDS ranking—mean accuracy drop from stride=1 to 16 averaged over all 8 architectures ($n=8$). FineGym (55.9pp) and AUTSL (53.0pp) are co-equal highest-demand domains (order between them is leave-one-out sensitive; see supplementary material); UCF-101 (4.9pp) is appearance-dominated. Positions 3–8 are exactly stable under leave-one-architecture-out. **Right:** Pearson correlation between optical-flow magnitude and per-class aliasing loss (8 datasets). AUTSL and DriveAct show the expected positive correlation; EPIC-Kitchens is near-zero (background camera motion confounds flow); FineGym shows the strongest correlation of any dataset but in the opposite direction ($r = -0.475$): discrete-event classes alias more than sustained-motion ones.

4.2 Cross-Architecture Temporal Aliasing

Attention type correlates with aliasing. Table 3 shows stride=1 accuracy and stride=16 drop. The central finding is that aliasing tracks *attention mechanism type* more closely than architecture family. The robust group is TimeSformer (divided attention, 10.3pp mean drop) and VideoMamba (SSM, 11.4pp). Both preserve global temporal context under aggressive subsampling: TimeSformer via full-clip cross-frame attention, and VideoMamba via recurrent state accumulation.

VideoMamba’s robustness is markedly *bimodal*. It drops only 4.0pp on average over UCF-101, HMDB-51, EPIC-Kitchens, Diving-48, and DriveAct, which would make it the most robust model by a large margin if these were the only domains considered. But on the three highest-TDS domains, VideoMamba suffers substantially: FineGym (40.8pp), AUTSL (16.0pp), and SSv2 (14.1pp). FineGym most clearly inverts the SSM-vs-Transformer ranking: TimeSformer drops 36.1pp vs. VideoMamba’s 40.8pp, a 4.7pp gap favoring divided attention. The same direction holds on SSv2, while AUTSL is tied. We hypothesize that VideoMamba’s running state smooths over high-frequency phase transitions, whereas TimeSformer can selectively attend to the most discriminative frames.

The SSv2 result extends beyond temporal frequency: SSv2 tests causal ordering (e.g., move object left then right), not just temporal density. SSM recurrence over skipped frames integrates corrupted order evidence, while appearance-driven UCF-101 remains stable. The fragile group comprises SlowFast (42.1pp), VideoMAE (32.8pp), ViViT (30.0pp), R2+1D (30.0pp), R3D-18 (29.7pp), and

Fig 1: Temporal Aliasing Curves — Accuracy vs Sampling Stride (coverage=100%)

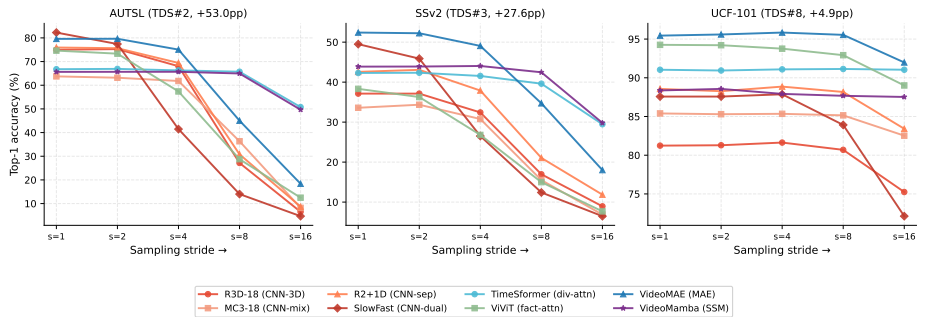


Fig. 4: Accuracy vs. stride (cov=100%) for three representative datasets. **AUTSL** (TDS=53.0pp): TimeSformer and VideoMamba remain most stable; all other models cliff above stride=4. **SSv2** (TDS=27.6pp): clear two-tier separation between robust (TSF, VMamba) and fragile models. **UCF-101** (TDS=4.9pp): minimal sensitivity across all architectures. These three datasets cover the full TDS range, validating the metric.

Table 3: Full temporal aliasing table: stride=1 accuracy (%) / drop at stride=16 (pp), cov=100%. Rows sorted by average drop across all 8 datasets.

Model (family)	UCF-101	SSv2	HMDB-51	Diving-48	FineGym	AUTSL	DriveAct	EPIC-Kit	Avg drop
TimeSformer (div. attn)	91.0/0.0	42.3/12.8	79.9/4.0	38.2/2.0	66.4/36.1	66.8/16.0	67.6/8.5	32.3/2.9	10.3pp
VideoMamba (SSM)	88.4/ +0.8	43.9/14.1	69.8/3.6	36.5/4.9	72.7/40.8	65.6/16.0	57.6/9.6	28.3/1.2	11.4pp
MC3-18 (CNN-mix)	85.4/2.9	33.6/26.7	78.6/9.7	31.6/12.4	64.2/51.1	63.7/55.5	69.0/13.8	36.2/8.6	22.6pp
R3D-18 (CNN-3D)	81.2/6.0	37.1/28.2	80.3/21.1	28.8/16.0	71.5/61.4	75.0/68.5	68.3/23.9	37.2/12.7	29.7pp
R2+1D (CNN-sep.)	88.6/5.1	42.6/30.7	73.1/17.6	35.3/18.8	76.8/64.2	75.9/67.2	62.5/24.8	35.5/11.7	30.0pp
ViViT (fact. attn)	94.3/5.2	38.3/30.6	80.2/19.4	53.0/36.3	69.9/55.2	74.6/62.1	67.4/20.3	32.9/11.1	30.0pp
VideoMAE (MAE)	95.4/3.5	52.4/34.4	84.0/20.2	48.6/23.4	78.0/67.8	79.6/61.2	73.9/40.8	37.7/10.8	32.8pp
SlowFast (dual)	87.6/15.4	49.5/43.0	79.3/37.4	50.5/39.7	78.2/71.5	82.3/77.6	72.5/33.7	39.4/18.5	42.1pp

MC3-18 (22.6pp). ViViT is the key anomaly: a Transformer whose factorized attention remains vulnerable to frame dropout.

Architecture-specific failure modes. SlowFast cliffs at stride 2→4 because its fast and slow pathways are simultaneously starved; on FineGym it has the best base accuracy (78.2%) but the largest drop (71.5pp). VideoMAE is spatially strong but temporally fragile: masked patch reconstruction does not train robustness to missing frames. VideoMamba’s failures concentrate on high-frequency phase changes (FineGym), handshape transitions (AUTSL), and ordering-sensitive interactions (SSv2).

Cross-architecture sensitivity heatmap. Figure 5 shows the full coverage×stride sensitivity matrix averaged over 8 datasets, with models sorted by mean TDS. The diagonal structure—robust models maintaining a flat plateau while fragile models show sharp gradients along the stride axis—is consistent across datasets (full per-model, per-dataset heatmaps are in the supplementary material).

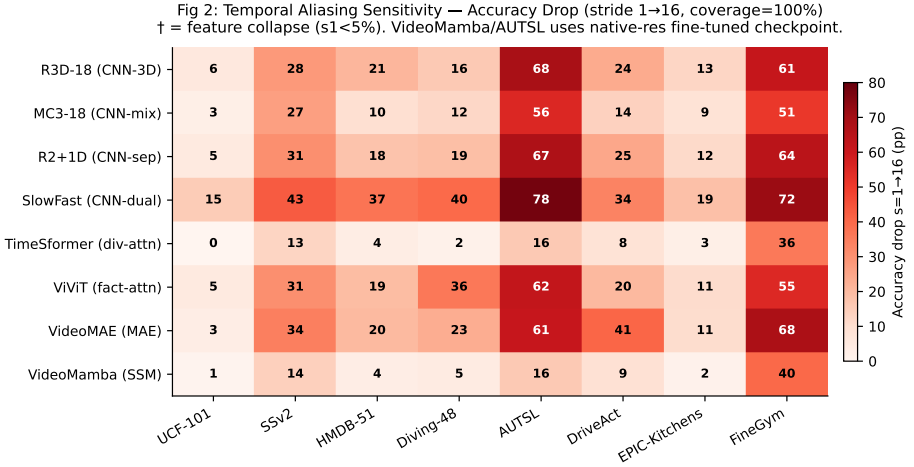


Fig. 5: Cross-architecture sensitivity heatmap: accuracy (%) averaged over 8 datasets across the coverage×stride grid. Models sorted left-to-right by mean aliasing robustness. VideoMamba and TimeSformer maintain a flat high-accuracy plateau; SlowFast and ViViT show dramatic stride-driven collapse (red gradient). VideoMamba/AUTSL uses the AUTSL-fine-tuned checkpoint, consistent with all other models’ AUTSL checkpoints.

Table 4: ANOVA effect sizes (η^2 , mean±std over 8 datasets). All stride effects significant ($p<0.05$). Coverage dominates; stride separates robust from fragile.

Model	η^2_{stride}	η^2_{cov}	Interpretation
VideoMamba	0.09 ± 0.14	0.88 ± 0.07	Robust
TimeSformer	0.11 ± 0.13	0.88 ± 0.06	Robust
MC3-18	0.20 ± 0.07	0.73 ± 0.10	Moderate
R2+1D	0.23 ± 0.08	0.69 ± 0.09	Fragile
R3D-18	0.24 ± 0.06	0.68 ± 0.07	Fragile
VideoMAE	0.20 ± 0.11	0.73 ± 0.14	Moderate
ViViT	0.26 ± 0.05	0.65 ± 0.09	Fragile
SlowFast	0.35 ± 0.07	0.53 ± 0.09	Most fragile

4.3 Statistical Validation (ANOVA and Levene’s Test)

Effect sizes. Table 4 reports two-way ANOVA η^2 values (mean±std over 8 datasets). Coverage is the dominant factor across all models ($\eta^2_{\text{cov}} = 0.53\text{--}0.90$). Stride effect size distinguishes the two groups: VideoMamba and TimeSformer have $\eta^2_{\text{stride}} \approx 0.08$, while SlowFast reaches 0.35—a 4.4× difference confirming that stride is architecturally consequential for fragile models. Full per-dataset η^2 tables are provided in the supplementary material.

Variance inflation. Levene’s test shows significant inter-class variance inflation in 67% of model-dataset pairs ($p < 0.05$, Bonferroni-corrected across all 64 model×dataset comparisons). The worst cases are VideoMAE/HMDB-51 (2.0×) and R3D-18/HMDB-51 (1.85×), while TimeSformer and VideoMamba remain

Table 5: Taxonomy tier boundaries (accuracy drop pp): classes below Low threshold alias minimally; above High threshold alias severely. Computed at cov=100%, stride=1→16.

Dataset	Low $\leq$	High $>$	# High	# Low
AUTSL	66.9pp	73.9pp	75	74
Diving-48	25.0pp	47.7pp	16	16
SSv2	59.7pp	76.9pp	58	57
HMDB-51	8.4pp	28.5pp	17	17
EPIC-Kitchens	0.0pp	19.4pp	30	11
DriveAct	14.8pp	51.0pp	11	11
UCF-101	1.4pp	7.5pp	35	31

stable ($\approx 1.1\times$). Thus sparse sampling can create class-specific failures that mean accuracy hides.

4.4 Action Sensitivity Taxonomy

We classify each action class into three tiers by per-class aliasing loss (accuracy drop, stride=1→16). Tier boundaries use the 33rd/67th percentiles.

High-sensitivity actions are temporally critical: in AUTSL, even the Low tier loses 38pp, confirming that the entire dataset requires dense sampling. In contrast, the UCF-101 Low tier loses only 0.3pp: static, appearance-driven actions are virtually unaffected. Table 5 reports per-dataset tier boundaries. AUTSL’s Low threshold starts at 66.9pp, above the High threshold of any other dataset, confirming that sign language forms a distinct temporal-frequency regime.

Clip duration effect. Counter-intuitively, shorter clips alias more severely (Pearson $r = -0.3$ to -0.8 across models): short clips have less temporal redundancy, so each missed frame removes a larger fraction of available evidence. On UCF-101, clips $>6s$ lose only 1.5pp at stride=16 vs. 16.2pp for clips $<3s$. This suggests that short clips should receive dense sampling irrespective of model confidence (details in the supplementary material).

4.5 Spatial Resolution and Target-Resolution Fine-Tuning

Cross-resolution robustness. Figure 6 and Table 6 show accuracy vs. input resolution for all 8 models on SSv2, evaluated *without retraining*. The same attention-type split observed in the temporal domain reappears here. CNNs (native 112px) degrade sharply at higher resolutions: R2+1D drops from 42.6% at 112px to 5.9% at 336px ($-36.6pp$). Transformers and VideoMamba (native 224px) stay within $\pm 7pp$ across 96–336px.

The CNN/Transformer split holds across all 8 datasets (supplementary material). On AUTSL, R3D-18 drops from 75.0% to 2.0% at 336px ($-73.0pp$), whereas Transformers remain within $\pm 8pp$ on 7 of 8 datasets; EPIC-Kitchens is the main exception.

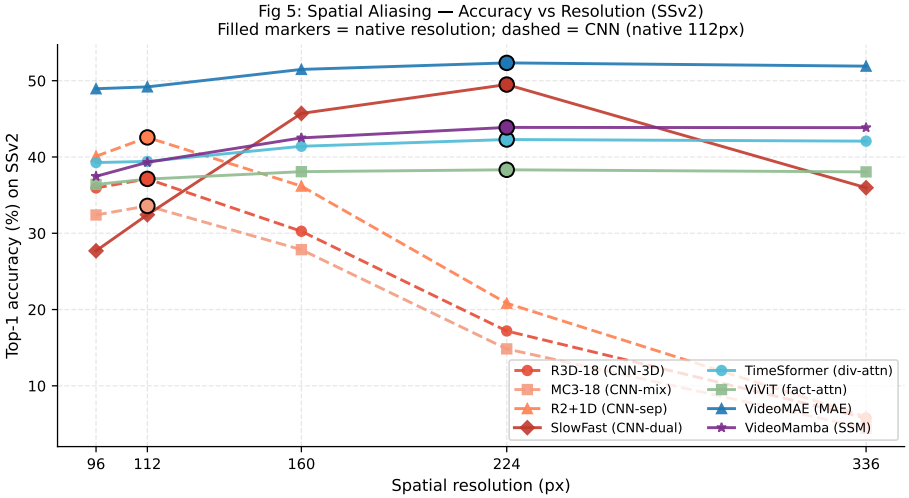


Fig. 6: Accuracy vs. input resolution on SSv2 (no retraining). CNNs (dashed, native=112px) degrade severely at higher resolutions; Transformers and VideoMamba (solid, native=224px) stay within ± 7 pp across 96–336px. Filled markers denote each model’s native training resolution.

Table 6: Spatial robustness on SSv2 (no retraining). Bold = native resolution. $\Delta_{\max}$ = largest drop from native.

Model	Native	96px	112px	160px	224px	336px	$\Delta_{\max}$
R3D-18	112px	35.9	37.1	30.3	17.2	5.8	−31.4pp
MC3-18	112px	32.4	33.6	27.8	14.8	4.4	−29.2pp
R2+1D	112px	40.1	42.6	36.2	20.8	5.9	−36.6pp
SlowFast	224px	27.7	32.4	45.7	49.5	36.0	−21.8pp
TimeSformer	224px	39.3	39.4	41.4	42.3	42.1	−3.0pp
ViViT	224px	36.4	37.1	38.1	38.3	38.0	−1.9pp
VideoMAE	224px	48.9	49.2	51.5	52.3	51.9	−3.4pp
VideoMamba	224px	37.5	39.3	42.5	43.9	43.8	−6.4pp

Target-resolution fine-tuning recovers CNN performance. Table 7 shows family-level averages after target-resolution tuning. CNNs recover sharply: at 224px, cross-resolution accuracy is 30.0%, but fine-tuning reaches 69.2% (+39.2pp). Transformers improve modestly because they are already robust across resolutions.

4.6 Inference Latency and Deployment Budgeting

Table 8 reports measured bfloat16 latency at batch 1. VideoMamba is slowest (8–15ms), 4–8 $\times$ slower than VideoMAE (1.6–2.8ms) despite its SSM design. VideoMAE gives the best Transformer accuracy per ms, while CNN latency scales predictably with resolution.

Table 7: Target-resolution fine-tuning: family-level mean accuracy (%) averaged over all 8 datasets, after fine-tuning from the native-resolution checkpoint for 10 epochs at each target spatial resolution. CNNs benefit by up to 39.2pp at off-native resolutions. Transformers already achieve near-native accuracy without retraining.

Family	48px	96px	112px	160px	224px
CNN (3 models)	49.3	64.9	67.7	68.5	69.2
Transformer (3)	36.6	56.3	60.3	66.9	71.4
SSM (VideoMamba)	31.5	41.2	45.1	54.5	59.7
Dual-CNN (SlowFast)	36.2	73.3	71.9	71.6	67.5

Table 8: Inference latency (ms/clip, bfloat16, batch=1, RTX PRO 6000 Blackwell). VideoMamba is 4-8× slower than VideoMAE despite being an SSM. Scale to other devices: RTX 4090 ×1.5, Jetson AGX Orin ×10, Jetson Nano ×50.

Model	48px	96px	112px	160px	224px
R3D-18	0.61	0.89	1.20	1.76	2.78
MC3-18	0.61	0.85	1.18	1.84	3.15
R2+1D	1.39	2.21	2.54	4.10	5.77
SlowFast	3.09	3.11	3.11	3.13	4.10
TimeSformer	3.25	3.24	3.20	3.20	4.30
ViViT	1.55	1.77	1.99	3.19	6.35
VideoMAE	1.57	1.79	1.78	1.92	2.82
VideoMamba	8.06	8.15	8.18	10.03	15.44

4.7 Entropy-Based Adaptive Routing

Table 9 compares entropy routing at average frame budget ≤ 8 (full curves in the supplementary material). AdaFocus [36] and AR-Net [20] use ResNet-50 and are informational. The controlled comparison is FrameExit [9], also on TimeSformer: entropy routing reaches 42.5% vs. 38.4% (+4.2pp) with fewer frames (7.7 vs. 9.9).

Across model-dataset pairs, 77% of videos route to cheap 4-frame inference on average. VideoMAE gives the highest absolute routed accuracy, while TimeSformer achieves the highest cheap-routing rates on temporal-critical datasets.

5 Discussion

Why is attention type associated with aliasing robustness? Divided attention (TimeSformer) computes temporal context globally at each block: every frame attends to all others, selectively weighting discriminative instants and reducing the impact of any single missing frame. *Factorized attention* (ViViT) extracts spatial features frame-independently before temporal aggregation; when frames are dropped, the temporal module receives impoverished per-frame representations it cannot recover from. *SSM recurrence* (VideoMamba) integrates temporal context as a running state: on slowly varying domains, missing frames perturb the state boundedly, but high-frequency key-frame evidence can be smoothed away.

Fig 4: E7 Entropy Routing — Accuracy vs Avg Frames (4f–16f cascade)

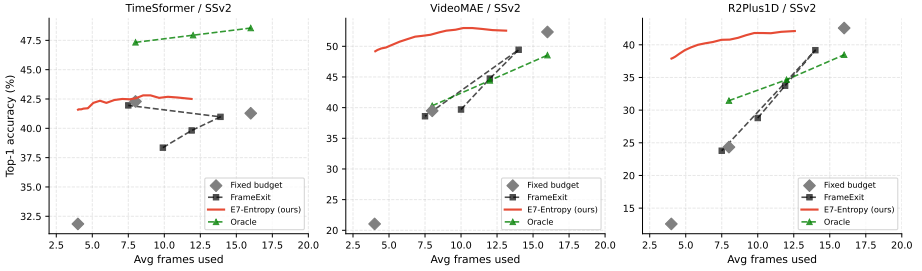


Fig. 7: Entropy routing: accuracy vs. average frames on SSv2. Each curve sweeps threshold $\tau \in [0.10, 0.95]$ between 4-frame cheap inference and 16-frame dense inference. VideoMAE and TimeSformer route most videos cheaply with no accuracy penalty.

Table 9: Routing comparison on SSv2 at avg ≤ 8 frames. The fair, same-backbone comparison is Entropy (ours) vs. FrameExit, both on TimeSformer. AdaFocus/AR-Net use a different backbone (ResNet-50) and are reported only as context for absolute numbers reported in their respective papers, not as a controlled comparison.

Method	Avg frames	SSv2 Top-1	Backbone
<i>Informational only: different backbone</i>			
AdaFocus [36]	8.0	54.7%	ResNet-50
AR-Net [20]	8.0	48.6%	ResNet-50
<i>Controlled comparison: same TimeSformer backbone</i>			
Oracle upper bound	7.7	47.3%	TimeSformer
Fixed 16f (dense)	16.0	42.3%	TimeSformer
Fixed 8f	8.0	42.3%	TimeSformer
Entropy (ours)	7.7	42.5%	TimeSformer
Fixed 4f (cheap)	4.0	41.5%	TimeSformer
FrameExit [9]	9.9	38.4%	TimeSformer

Full characterization across all 8 datasets reveals VideoMamba’s **bimodal** aliasing profile: robust on 5 datasets (avg 4.0pp), but substantially aliased on FineGym (40.8pp), AUTSL (16.0pp), and SSv2 (14.1pp). These domains expose three SSM failure modes: bandwidth limits for fine-grained phase changes, ordering errors under subsampling, and loss of spatial-temporal precision.

Spatial robustness mirrors temporal robustness. Cross-resolution results show that the architectural property governing temporal aliasing also relates to spatial scale sensitivity. CNNs’ strided convolutions learn features at a specific spatial scale; off-resolution, spatial frequency patterns shift relative to learned filters, causing catastrophic accuracy loss. Global attention aggregates tokens across scales, giving Transformers uniform robustness (≤ 8 pp on 7 of 8 datasets). VideoMamba again behaves bimodally, matching Transformers on some datasets but degrading on cluttered egocentric and in-vehicle scenes.

The ViViT result is particularly instructive: despite being temporally fragile, ViViT is spatially robust. This confirms that patch tokenization followed by attention is the source of spatial invariance, independent of whether temporal

attention is factorized or divided. Temporal and spatial robustness therefore arise from related but distinct architectural properties.

Joint budgeting. Resolution, latency, and sampling density should be optimized jointly. Target-resolution fine-tuning turns resolution into a cost dial: retrained CNNs at 96px average 64.9% at 0.89ms, while Transformer 224px inference averages 71.4% at 2.82ms, a $3.2\times$ latency reduction for a 6.5pp accuracy cost.

On the temporal axis, a symmetric convergence occurs: temporally robust architectures at aggressive stride can match fragile ones at dense stride. R2+1D at stride=1 achieves 42.6% on SSv2; TimeSformer with entropy routing at avg 7.7 frames reaches 42.5%, essentially identical accuracy at less than half the frames. Practitioners should therefore search jointly over architecture, resolution, and sampling density rather than treating them as independent axes.

Nyquist-aware deployment. From TDS and measured latency, we derive deployment rules. **High-TDS domains** (FineGym, AUTSL; TDS > 40pp) require stride ≤ 2 ; **moderate-TDS domains** (SSv2, DriveAct) tolerate stride ≤ 4 ; and **low-TDS domains** (UCF-101, EPIC-Kitchens) tolerate stride up to 16. Transformers usually work across 48–224px without retraining, while CNNs should be fine-tuned at the target resolution. ViViT’s fragility shows that “Transformer” does not imply temporal robustness, and VideoMamba’s 4–8 $\times$ CUDA latency gap shows that theoretical SSM efficiency does not automatically translate to best GPU accuracy-per-ms.

Limitations. Our 8-architecture pool covers major temporal-sampling families but not newer hybrid models such as Video Swin, MViT, or InternVideo-style systems. The Nyquist–Shannon framing is conceptual: our direct whole-frame-flow test (Sec. 4.1; supplementary material) finds an inverse, not positive, relationship with stride-sensitivity, so we characterize aliasing through controlled accuracy degradation rather than claiming literal spectral validation. Flow and clip-duration analyses remain exploratory, and routing uses a global 4f/16f threshold; per-dataset calibration could close part of the 4.8pp oracle gap.

6 Conclusion

We presented a large-scale cross-architecture study of spatiotemporal aliasing in video action recognition. Across 8 architectures, 8 datasets, and 8,000+ configurations, aliasing tracks attention mechanism type more than broad model family, with the ViViT anomaly demonstrating that Transformer $\neq$ temporally robust. The TDS metric identifies FineGym and AUTSL as co-equal high-demand domains, establishing fine-grained sport as a high-aliasing regime. VideoMamba is robust but bimodal and slow on GPU; target-resolution fine-tuning closes the CNN resolution gap by up to 39pp; and entropy routing cuts inference cost by 52% without accuracy loss. An anonymized dashboard/code package accompanies the submission, with public release planned after review. Together, these

results give practical rules for architecture, resolution, and stride selection under sampling constraints.

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